EECE 5141/6041C: MICROFABRICATION LAB FOR SEMICONDUCTOR DEVICES AND MEMS

FINAL LAB REPORT ON FABRICATING MEMS PRESSURE SENSOR

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INTRODUCTION

The following report details the steps undertaken to produce and test a micro electromechanical system (MEMS) based silicon microchip capable of sensing pressure. A bare silicon wafer was used as the starting material, from which a series of masking, etching, and metallization, bonding, among other techniques, were used to create pressure sensors in the silicon. After fabrication, a variable pressure chamber was used to test one such MEMS pressure device to determine a calibration curve for it. It was found that the device under test (DUT) modulate the output signal in response to changes in pressure.

LAB 1: BAKER CLEANING PROCESS AND FIELD OXIDATION

AIM:

To clean and oxidize the bare Silicon wafer.

For designing and fabricating pressure sensor for this lab course, we are using 2-inch n-type Silicon wafer (100) orientation with <110> flat and substrate: n-type and 1~10 Ω.cm.

THEORY:

The bare Si wafer we get from the manufacturer is not clean and lot of contaminants are present. And also, many fingerprints are also present. So before using the wafer we need to ensure that the wafer is clean and hence the Baker Cleaning process is carried out. After the wafer is cleaned, we need to oxidize the Si wafer in order to get the silicon dioxide layer on the wafer. While holding the wafer with tweezer, hold the flat side with the tweezer.

Hydrofluoric acid (HF) is used to removed native oxide. It removes around 1000Å of the native oxide present on the wafer. JTB-100 is used to remove particles, metals and organic contaminants present on the wafer. While heating the solution of JTB-100, make sure to cover the Pyrex beaker with aluminum foil and place thermometer into the beaker to keep an eye on temperature. As HF is dangerous, so handle it with extreme caution and always use a Teflon beaker and wear apron, gloves and safety shield. As HF fumes are fatal always do all the processes on the wet bench with exhaust or in the Fume Hood. While pouring HF into the beaker, do not pour it over the sink. Pour extra HF into hazardous waste of HF can. Never leave it unattended. In case you want it for future use, make sure to label it and provide your details so it is easier for the other person.

PROCEDURE:

1. Baker Cleaning Process

   First off, clean the Teflon beaker with distilled (DI) water. Mix 50 ml 49% HF into DI water in a Teflon beaker (Always add HF into DI water and not DI water into

HF). Dip the wafer using a Teflon holder for 1minute. Then rinse the wafer under flowing DI water for about 3 minutes.
Now, to remove particles, metals and organic contaminants, JTB-100 is used. For this, mix 120ml of JTB-100 into 600ml DI water in a Pyrex glass beaker. Heat the solution up to 70°C on a heat plate. Pour 24ml H_2O_2 when the solution is heated up. Dip the wafer into the solution for 10 minutes. While the wafer is being dipped, keep an eye on the temperature as it should not be too high or too low. Then, rinse the wafer under flowing DI water for 3 minutes and dry the wafer using N_2 gas gun until all the water is removed on both sides of the wafer (While using N_2 gun, always put wipes below the wafer and the flow of the gun should be towards the wafer). Keep changing the position and angle of the wafer while drying so residue is left.

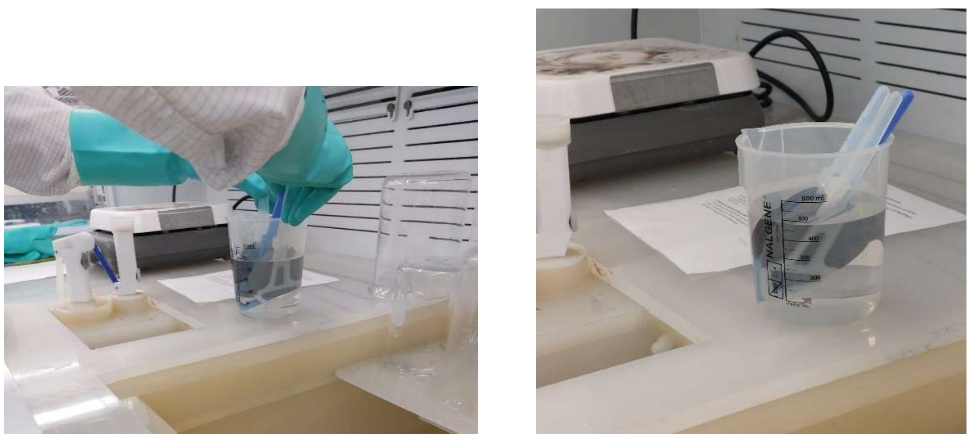


Figure 1 & 2: Wafer dipped in HF solution

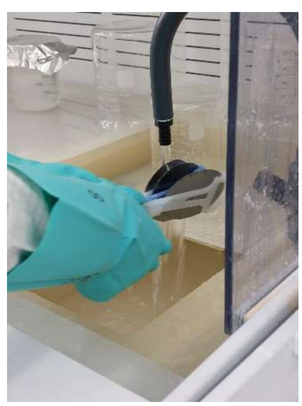


Figure 3: Rinsing wafer under flowing DI water

Figure 4: Heating JTB-100 solution

Figure 5: Drying wafer with N_2 gun

2. Field Oxidation

   For oxidation, dry oxidation results into the best quality but as its oxidation rate is very less, only thin oxide layer is obtained even after loading it for longer period of time. Wet oxidation yields into thick oxide layer in shorter duration but the quality is not that good. So, in order to achieve good quality thick layer of silicon oxide, dry, wet and dry oxidation process is carried out.

   Load the wafer in N_2 gas chamber at 1000°C. First perform dry oxidation for 30 minutes followed by wet oxidation for 5.5 hours and then again followed by dry oxidation for 30 minutes. After oxidation is done, unload the wafer. During the reaction, pay more attention to the center position temperature because wafers are put in the center of the furnace.

LAB 2: PHOTOLITHOGRAPHY OF ANISOTROPIC ETCHING MASK FOR SI MEMBRANE

AIM:

To measure the thickness of oxide that we obtained from our previous lab and performing photolithography of anisotropic etching for Si membrane.

THEORY:

To measure the grown thickness of the silicon oxide layer, Ellipsometer is used. To know whether Ellipsometer is working is correctly or not, test it with a bare Si wafer. The native oxide on the bare SI wafer is not more than 100Å. Ellipsometer shows many different types of readings so choose whatever readings you want to observe. Here, we need to measure the thickness of grown silicon oxide so choose that. Measure thickness on both sides of the wafer. In order to get accurate readings, always take three readings and then do the average. To measure, press "measurement" on keyboard. To accept, press "yes" otherwise "no".

For anisotropic etching using TMAH, for the desired dimensions of the etch, mask is used to expose that area and then using photolithography this is achieved. Here, we are using positive photoresist. To expose the desired region, wafer is put underneath the alignment mask and then adjusted to get the accurate outcome. The distance between wafer and mask should not be too less otherwise it would lead to bad exposure. After the mask is aligned, UV exposure is carried out. After it is exposed, the wafer is observed under the microscope to check whether the exposure is proper or not. It should ne over-exposed or under-exposed, this would result into poor quality. Here, PR used is AZ1518, 15 describes 1500 series and 18 describes the thickness available in µm in spinning the wafer at 4000rpm.

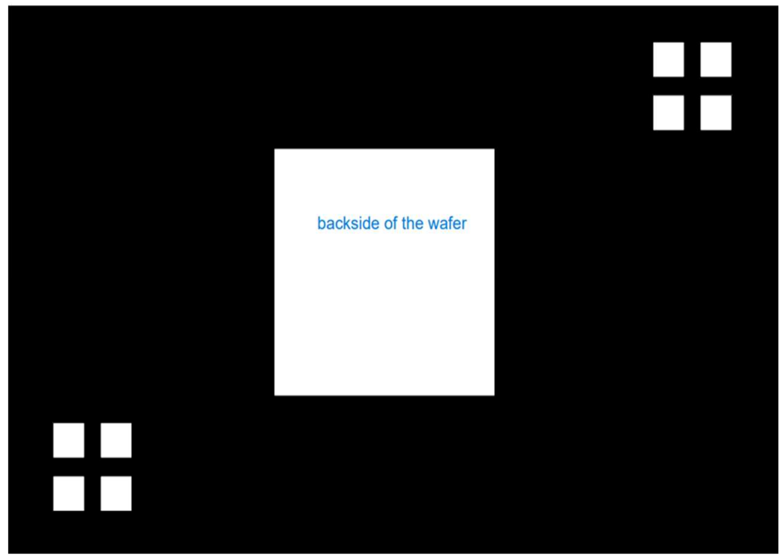


Figure 6: Alignment mask

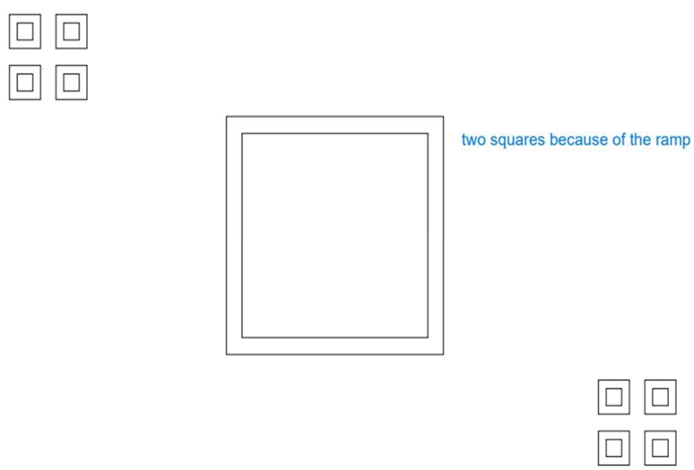


Figure 7: After exposure

PROCEDURE:

For measuring thickness of grown oxide layer, Ellipsometer is used. Three readings on each side is taken in order to achieve accurate readings because more the readings, better the accuracy.

Theoretical Calculation:

Oxide Thickness Calculation: Using Deal Grove Rate Constants

At 1000 C. Dry 30 min, A=0.165μm and B = 0.0117μm ; τ = 0.37 hr

Xf= $0.5 \times 0.165(\sqrt{1 + \frac{4\times0.0117}{(0.165)^2}(0.5 + 0.37)} - 1 = 0.047\mu m$

Wet 5.5 hrs, A =0.226 μm. B=0.287μm and ; τ =$\frac{0.047^2+0.226\times0.047}{0.287} = 0.044\ hr$

Xf = $0.5 \times 0.226(\sqrt{1 + \frac{4\times0.287}{0.226^2}(0.044 + 5.5)} - 1 = 1.15\mu m$

Dry 30 min, A=0.165μm and B = 0.0117μm ; τ = $\frac{1.15^2+0.165\times1.15}{0.0117}$= 129.25 hr

Xf =$0.5 \times 0.165(\sqrt{1 + \frac{4\times0.0117}{0.165^2}(129.25 + 0.5)} - 1 = 1.152\mu m$

Final Thickness = 1.152μm

Readings for wafer #1-

Back Side:

Readings	Oxide Thickness(Å)	Average thickness(Å)
R1	10945	10955
R2	10953	
R3	10967	

Front Side:

Readings	Oxide Thickness(Å)	Average thickness(Å)
R1	10996	10992
R2	10988	
R3	10993	

Readings for wafer #7 (Practice Measurement)-

Back Side:

Readings	Oxide Thickness(Å)	Average thickness(Å)
R1	10884	10884
R2	10887	
R3	10882	

Front Side:

Readings	Oxide Thickness(Å)	Average thickness(Å)
R1	10882	10870
R2	10850	
R3	10877	

Figure 8: Ellipsometer

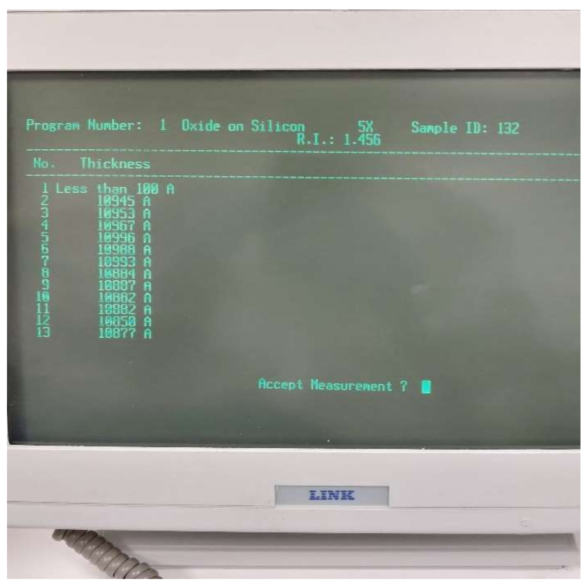


Figure 9: Screen with readings of Ellipsometer

For photolithography, first clean the wafer using acetone and then propan-2-ol (isopropyl alcohol) to remove organic solvents. Then rinse the wafer under flowing DI water for about 2 minutes and then dry it using N_2 gun.

After cleaning, coat the wafer with HMDS. HMDS is the adhesion promoter between Si wafer and photoresist. For coating, put several drops of HMDS onto the wafer to cover 2/3rd of the wafer and then spin-coat it using Recipe 27. While putting drops of HMDS make sure that there are no bubbles are present and the wafer is on the center of the spinner. This HMDS coat makes the surface hydrophobic so that the PR sticks better on to the wafer. After HMDS coating is done, perform similar steps with the PR. Here, we are using positive PR AZ1518. After PR is done, put on hot plate at 110°C for 5 minutes for soft bake. While putting wafer on to the hot plate for soft-bake, do not use the tweezer as the PR coating is fresh. After soft-bake is done, let it cool down for about 2 minutes.

Figure 10: Putting wafer on to spin coater

Figure 11: HMDS Coating

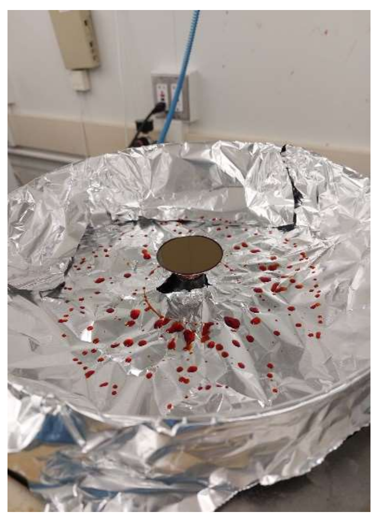


Figure 12: PR Coating

Figure 13: Soft-bake

For UV exposure, take the alignment mask. Clean the mask with DI water. Set the mask in the aligner. Put the wafer beneath the mask and then align it with the mask. The flat surface should be aligned with the line present on the mask. The micrometer screw gauge like tool is used to rotate and align the mask. The one beneath the gauge like tool (the black one) is used to bring wafer up or down (shift vertically). The one on the left of the black one is used to adjust wafer along Y-axis and the on the right side is used to adjust wafer along X-axis. The wafer should be visible form the windows of the mask. For perfect exposure, make sure that there is no space between mask and wafer. For this, observe the shadow of the wafer for reference. Now, measure the intensity of the UV light. It was found to be 6.86mW. To transfer a total amount of 100mJ of UV energy into the substrate, the exposure time should be 100/6.86 which is equivalent to 14.6 seconds. If it is 5.9mW set is for 15 seconds. If more or less, set accordingly. Here, we set it for 10 seconds and then observe it. If the exposure time is more, it leads to over-exposure so be careful about the time. DO NOT LOOK TOWARDS THE UV EXPOSURE WHEN IT IS TURNED ON. IT COULD DAMAGE YOUR EYES.

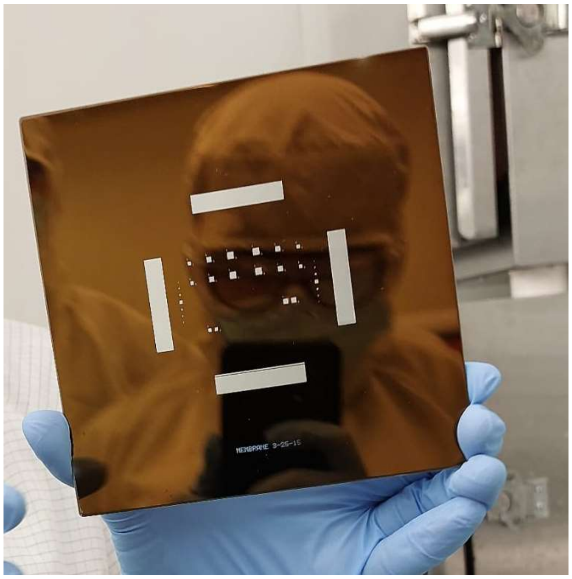


Figure 14: Alignment mask

Figure 15: Aligner and Exposer

After exposure, to remove exposed oxide area, wafer is dipped into the developer for 55 seconds and agitate the wafer. Developer is made mixing DI water and AZ400K in 3:1 ratio. Then rinse wafer under running DI water for 2 minutes and then dry it with N_2 gas. Observe under microscope. If bubbles are observed over the edges, then wash the wafer with acetone and iso-propyl alcohol and repeat the process. TO AVOID BUBBLES, REDUCE EXPOSURE TIME.

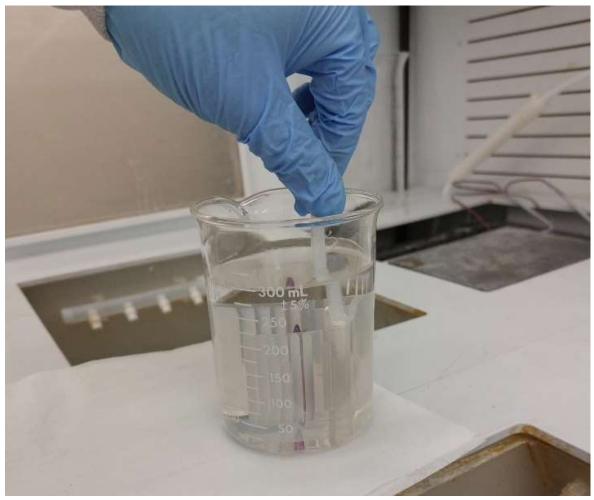


Figure 16: Agitating wafer in developer

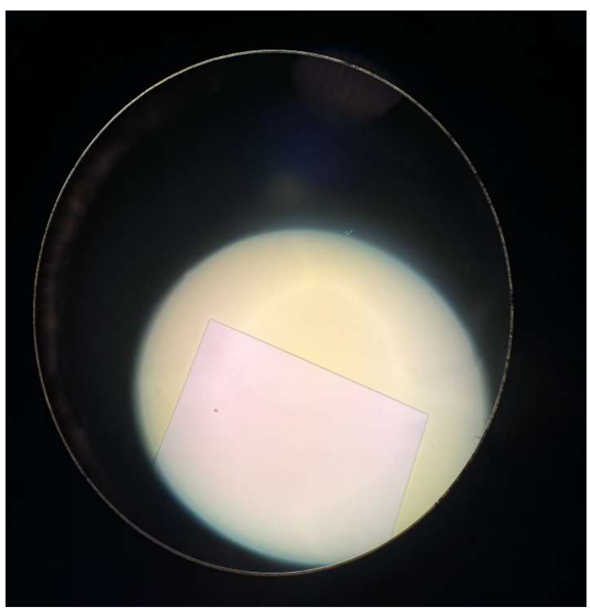


Figure 17: Observing under microscope

After observing the wafer under microscope, hard bake is done at 140°C for 10 minutes.

Figure 17: Hard bake

For protecting the back side of the wafer while etching, a dummy 3” wafer is used. First, the wafer is cleaned and coated with PR AZ1518 using spin coating. After this, place PR-coated 2” wafer that we are using on the top of 3” PR-coated dummy wafer. Using cotton swab, apply paint on the edge of 2” wafer to prevent possible etching on the edge. Also, cover the mask on the PR created by the tweezer. Leave it for drying overnight. Here, paint is used as adhesive.

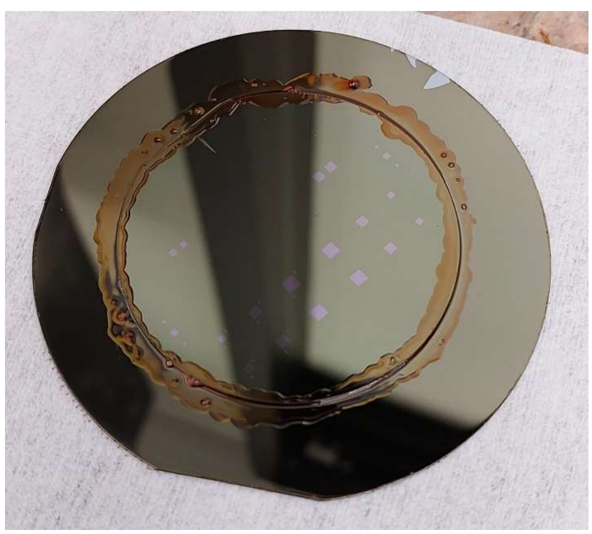


Figure 18: Sealing 2” wafer

LAB 3: BUFFERED OXIDE ETCHING (BOE) FOR ANISOTROPIC ETCHING OF Si DIAPHRAGM

AIM:

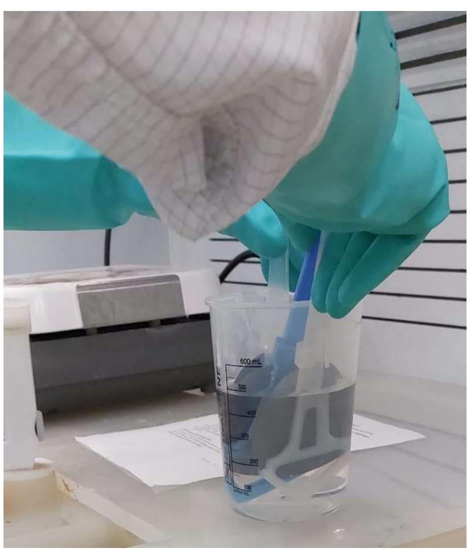
To perform BOE etching in order to have an opening for anisotropic etching of Si diaphragm.

THEORY:

The etch rate of oxide in BBOE solution is 0.5-1 µm/min at room temperature. We need to remove oxide layer of 10 µm so immerse wafer in BOE solution accordingly. NH_4F is used in BOE solution which is very toxic so handle with extreme caution. Wear gloves, apron and safety shield while using BOE solution. Pour extra NH_4F into HF hazardous waste can. Do not leave unattended.

PROCEDURE:

First, prepare BOE solution mixing 210ml DI water, 210ml 40%NH_4F and 70ml 49% HF in 3:3:1 ratio. Start with dipping the backside-protected wafer for 9 minutes in BOE solution and then observe under microscope. If we some oxide then performing etching for some more time. After the etching is done, rinse the wafer under running DI water for about 3 minutes and then dry the wafer using N_2 gas.



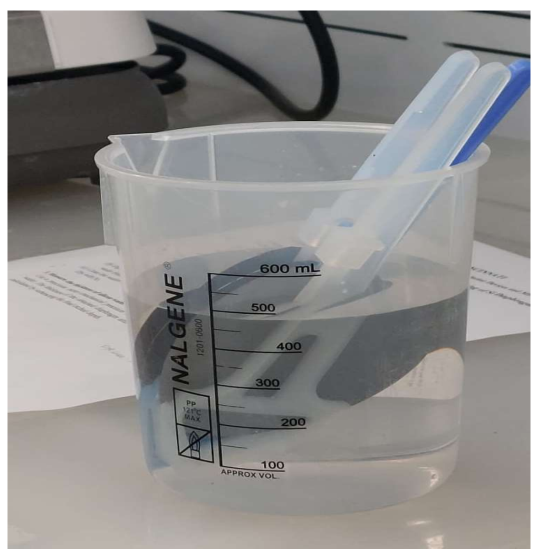


Figure 19 & 20: Wafer immersed in BOE solution

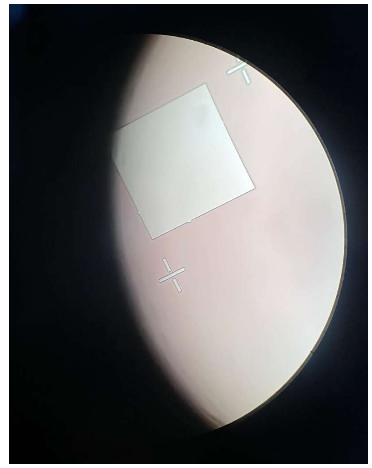


Figure 21 & 22: Observing under microscope

To remove photoresist, pour NMZ rinse into large glass beaker and heat it to 80°C covering it with aluminum foil. Place thermometer to keep an eye on the temperature. Dip wafer into the solution after it is heated for 10 minutes. This will separate both wafers. Here, as the PR is fresh, it takes less time otherwise it takes much more time to dissolve. NMZ rinse accelerates the dissolving of PR. Clean the wafer using acetone, IPA, DI water to make sure no PR residue is present. Then dry with N_2 gas.

After this, use precision meter to measure average thickness of the wafer.

Readings	Thickness of wafer(mm)	Average thickness(mm)
R1	0.277	0.280
R2	0.284	
R3	0.278	

LAB 4: ANISOTROPIC ETCHING FOR Si DIAPHRAGM FABRICATION

AIM:

To perform anisotropic etching using TMAH.

PROCEDURE:

Etch wafer again in dil. HF solution that we used in lab 1 to remove native oxide. To remove the residual silicon oxide film and contamination, after this dip, the wafer should be IMMEDIATELY put into the 25% TMAH solution without washing in DI water or drying out to prevent native oxide from forming again. The desire thickness of Si is 90μm and as we measured the thickness in the last lab, it is around 280 μm, the etch rate is 10 μm/hr so the total time would be around 10 hours. The etch rate will not be uniform throughout. As the PR layer is not fresh, it will reduce the etch rate. To remove stress during TMAH etching, second etching is done. TMAH etching is done in the oil tank(hood) as it will heat TMAH in a chamber and would improve etch rate. Measure thickness after every interval in order to calculate the remaining time. Check every 30 minutes after the thickness reaches to 100 μm. DI water is put next to TMAH solution to maintain temperature and condense the evaporated vapor. The temperature in the hood should be around 75°C. On measuring temperature, it was 66°C. After 16 hours, the Si thickness was 138μm. After TMAH etching, measure the thickness using precision meter.

Readings	Thickness of wafer(mm)	Average thickness(mm)
R1	0.113	0.138
R2	0.176	
R3	0.149	
R4	0.114	

After TMAH etching, dip the wafer in BOE solution prepared in Lab 3 to strip all the oxide before reoxidation. The thickness is 1 μm so 10 minutes BOE etching should be enough to strip all oxide.

Figure 23: TMAH hood

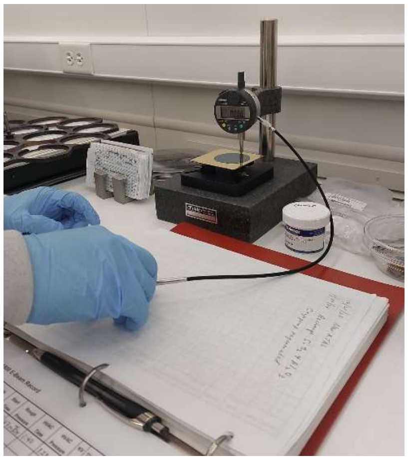


Figure 24: Measuring thickness using precision meter

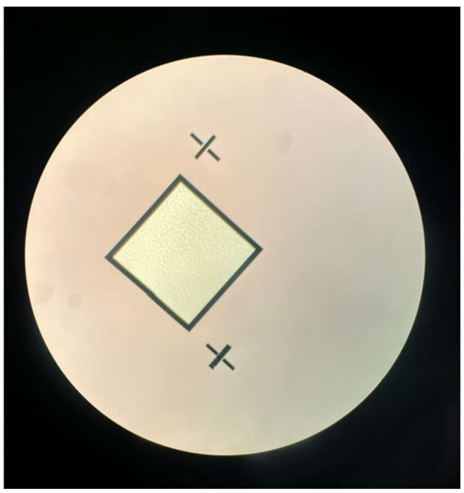


Figure 25: Observing under microscope

LAB 5: RE-OXIDATION AND PATTERNING OF PZR LINES FOR BORON IMPLANTATION

AIM:

Removing the oxide layer and re-oxidizing it as the quality is reduced because of etching and patterning PZR lines for boron implantation.

PROCEDURE:

The wafer was cleaned with the Baker cleaning process as done previously, after which it was oxidized at 1000°C. To do so, the wafer was introduced to the furnace under nitrogen, after which it was exposed to a dry oxygen atmosphere for 30 minutes, followed by a wet oxidation for 90 minutes, and a subsequent 30-minute dry oxidation (30-90-30 minute, dry-wet-dry). The oxide thickness was then measured by ellipsometer and averaged as done previously, results and average in Table R1. The wafer was then spin coated with resist as done previously and loaded into the mask aligner. From here, the infrared backside alignment feature on the mask aligner was used. The pattern for the piezo-resistors was exposed on the non-window (resistor) side of the wafer, using the backside feature to algin the resistor mask holes to the windows on the opposite side of the wafer. From here the mask was exposed, developed, and hard baked as done previously. An image of the alignment process and the final photoresist surface can be found in Figure R1, left and right respectively. The wafer was then etched using BOE as done previously, and thereafter the residual etchant removed. The wafer was then sent to an outside company for ion implantation.

Three attempts were made to align the mask before adequate alignment was achieved. While the exposure times of the photoresist were longer than expected based on the datasheet, when compared to previous experiments this was typical. The etching process appears to have adequately etched through to the silicon.

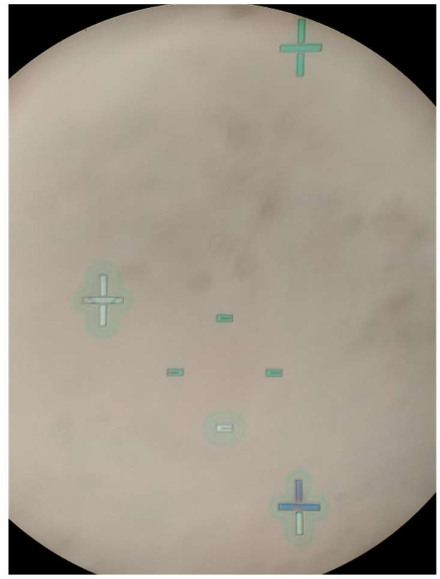
 

Figure 26 & 27: Shows the backside alignment screen (left) and the exposed photoresist surface (right).

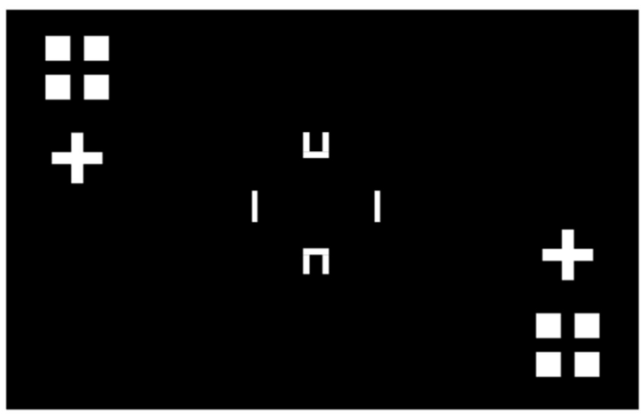


Figure 28: Shows the second photomask, providing the pattern to etch regions for doping for piezoresistors.

The measurements of the wafer's window and non-window sides following the pre-piezoresistor oxidation sequence are as below:

Readings	Window-side oxide thickness (Å)	Resistor-side oxide thickness (Å)
R1	6979	7178
R2	6896	7184
R3	6890	7159
Average	6922	7174

LAB 6: BORON ION IMPLANTATION AND ANNEALING/DRIVE-IN

AIM:

To perform annealing to heal the damaged tracks in the semi-conductor crystal lattice due to high energy implantation and also to activate the implanted boron.

PROCEDURE:

Ion implantation was carried out by an external partner at 50keV of 8x1014 boron atoms per cm^2, with a 7-degree tilt, a projected range of 1566Å, and a straggle of 504Å. The wafer was then stripped of photoresist using AZ NMP photoresist stripper at between 80°C to 90°C for 1 hour and cleaned with the Baker process as done before. Following this the wafer was annealed at 900°C in a nitrogen atmosphere for 30 minutes, after which a 10-40-10minute dry-wet-dry oxidation was performed and measured for thickness as described previously, results and average can be found in table. The resistor-window-side of the wafer was then coated with photoresist as done previously and hard baked. The BOE process was then used to remove the oxide from the window-side in preparation for anodic bonding. The PR was then stripped, and the wafer cleaned with the Baker process as done previously. The resistor- and window-side are shown in figure below (left and right, respectively). The implanted regions appeared as expected, and the wafer was clean and exhibited nice lines in the etched features following the cleaning steps indicating a successful transfer and alignment.

Readings	Window-side oxide thickness (Å)	Resistor-side oxide thickness (Å)
R1	3696	8066
R2	3710	8095
R3	3650	8070
Average	3685	8077

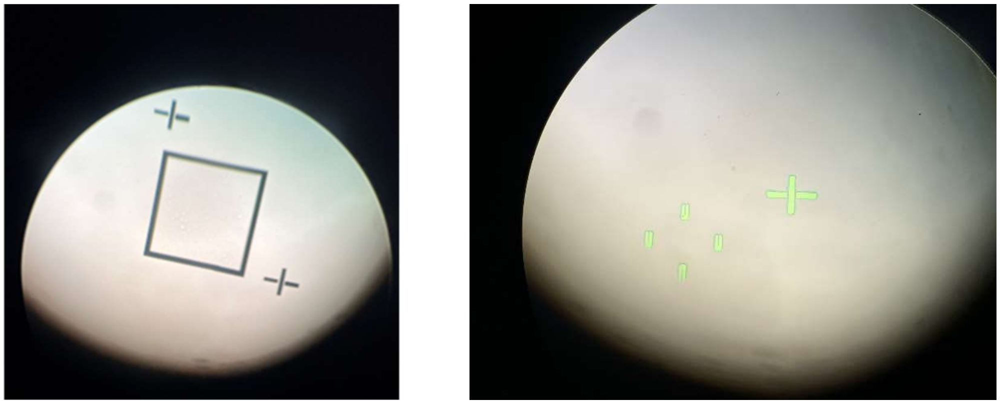


Figure 29 & 30: Shows the resistor-side post implantation and cleaning (left) and the window side (right).

LAB 7: SILICON-GLASS ANODIC BONDING

AIM:

To perform anodic bonding between Si and glass wafer.

PROCEDURE:

First, the silicon wafer and a second all-glass (Pyrex) wafer was cleaned using the Baker process. The silicon wafer was then dipped into a 2% HF solution for 60 seconds and then rinsed with DI water and dried with nitrogen. The anodic bonding heat chuck was set to heat at a rate of 15°C/minute to reach 450°C. At 200°C, the wafers were placed onto the chuck. The silicon wafer was then placed on the anodic bonding rig with the window-side face up, and the glass wafer placed on top of that surface, and allowed to reach 450°C. The contact point was then placed in the center of the wafer (final wafer-glass-needle assembly shown in Figure 31). The power was then turned on, a 10mA-limited current supply at 500-600 volts was used to bond the wafers. The minimum voltage required was used, 500V. The bond zone slowly spread from the contact point across the wafer until it nearly fully encompassed the surface, requiring 1.5. The bond progression can be seen in Figure 32. After bonding, the heat was turned off, the system allowed to cool, voltage turned off, and wafers removed. While some incomplete bonding regions were visible both at the edges and in the center (likely due to contamination), the overall bond was sufficient.

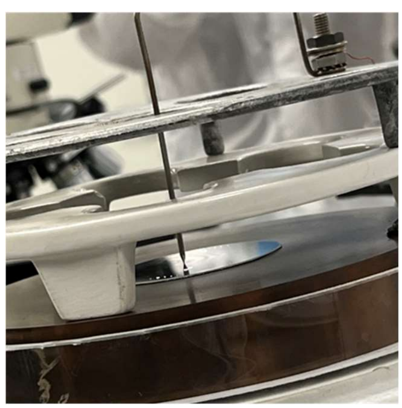


Figure 31: The wafer stack compressed under the anode point, sitting on the heat chuck.

Figure 32: Bond progression beginning at the anode point and spreading from it outwards radially until encompassing the entire wafer. Red arrows indicate regions of incomplete bonding at the edges, and center (possibly due to contamination).

LAB 8: 3RD LITHOGRAPHY (VIA OPENING)

AIM:

To open vias on the oxide layer at the top of the piezoresistors layer to make openings for the ohmic contacts.

PROCEDURE:

The wafer was cleaned using the Baker process as described previously. After, the resistor-side of the wafer was coated with photoresist as described in previous steps. The wafer was then exposed using mask 3 (Figure 34), developed, baked, and etched with BOE and cleaned as done previously. The photoresist was then removed from the wafer with acetone, isopropanol, and DI water to ensure complete resist removal. The resistor-side of the wafer exhibited bright square regions within the slightly-darker piezoresistor, indicating that the etching process proceeded successfully (Figure 33).

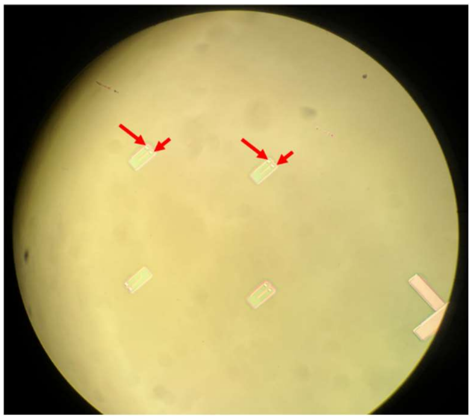


Figure 33: The vias etched in the resistor-side. The light regions pointed at by the red arrows are the holes etched to the piezo surface.

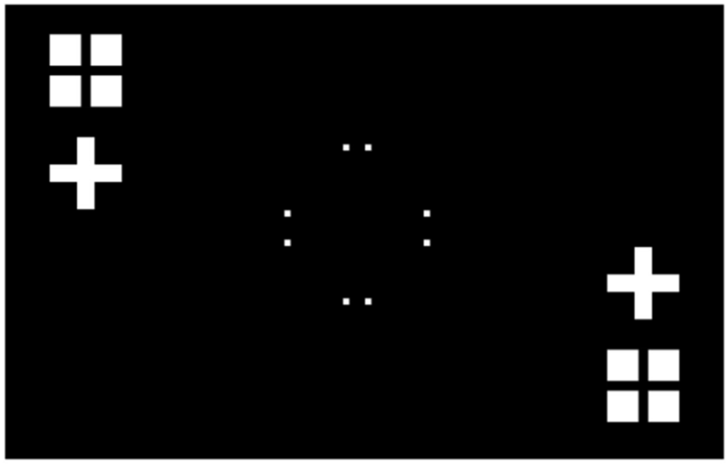


Figure 34: The mask used to pattern for etching of vias to the piezoresistors (Mask3).

LAB 9: ALUMINIUM DEPOSITION AND SINTERING

AIM:

To perform aluminum deposition by e-beam evaporation method and sintering.

THEORY:

Many materials such as metals, some semiconductor materials and insulators can be deposited as thin film by Physical Vapor Deposition method. 3 techniques of PVD can be used to deposit Al over the SiO2: Evaporation, Sputtering and Ion beam deposition. In this experiment we are using evaporation technique, more specifically e-beam evaporation technique.

The material to be deposited is placed in a crucible in the vacuum chamber. The material is heated till it melts and vaporizes using a high-energy electron beam. The e-beam is generated by running a current through a tungsten filament, which gets heated up and starts to emit electrons. These electrons are accelerated by applying a strong positive voltage between the filament and target material. The electron gun source is located under the crucible, so the e-beam is bent 270° by a permanent magnet. The e-beam collides with target material, transferring kinetic energy to the target atoms and significantly heating it up. The vaporized material travels through the vacuum and deposits over the substrate, forming a thin layer. The heating temperature depends on the vapor pressure of the material.

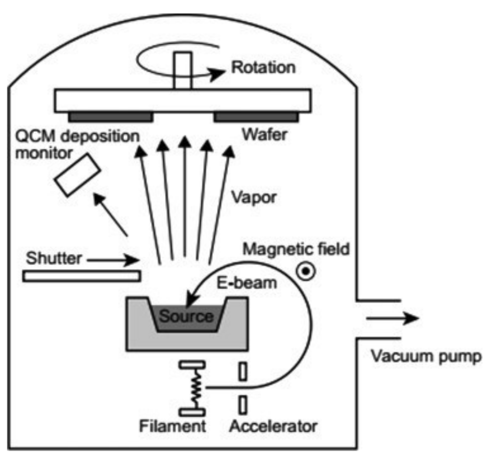


Figure 35: E-Beam Evaporation

Although Al adheres well to SiO2, it is challenging to form Al-S contact. Carrying out sintering/ annealing process after deposition is important. Annealing is done to achieve good ohmic contact and remove stress between the Al film and Si substrate. Typically sintering is carried out at temperatures around 450-500°C for 30 minutes.

PROCEDURE:

-First, we clean the wafer in 2% HF by 1 min dip, to remove the Native SiO2 and reduce its thickness as too much Al2O3 may make it difficult for Al to reach Si.

-Then aluminum source is loaded in the crucible in the vacuum chamber.

-The tungsten filament is connected to 9000V DC signal. Heat up the filament to evaporation temperature by slowly increasing the filament current.

-Once evaporation temperature is reached, remove the shutter

-The e-beam will start to heat up the target, it is a hit one spot continuously. The e-beam can be controlled to hit which spot we want.

-The deposition starts and once the desired thickness is deposited, close the shutter and power it down.

-The deposition rate: 10 A°/s; Emission Current is 200 mA, and the Desired thickness is 0.5μm

Vacuum Chamber:

> For longer mean free path. Evaporation temperature is lower on vacuum chamber
>
> Pressure: 10-3 - 10-4 Torr; Temperature in chamber: 10°K, gases usually solidify at this temperature, but Helium and Hydrogen gas is not frozen.

Aluminum Thickness monitoring:

> Acoustic impedance method- Gold plated crystal vibrates at 6 MHz It generates sound waves that travel through the wafer, a receiving transducer detects the reflected sound waves. It is compared with the original sound wave and the acoustic impedance is calculated. The acoustic impedance is used to determine the thickness of Al. We need to enter the density and z ratio of the source in the thickness monitor.

-After deposition, carry out thermal annealing/sintering in the oven at Temperature 500° C for 5 min

-During annealing we introduce nitrogen gas into the oven to prevent oxidation and to cooldown wafer after annealing.

Figure 36: Annealing/Sintering after Deposition

-Remove after it is below 200°C.

CALCULATION:

Deposition Rate: 10Å/s, desired thickness is 0.5μm

$$\text{Time required for deposition} = \frac{0.5\mu m}{10 A^{\circ}} = \frac{0.5 \times 10^{-6}}{10 \times 10^{-10}} = 500\text{ s} = 8.33\text{ mins}$$

LAB 10&11: ALUMINUM METAL LINE PATTERNING

AIM:

Patterning of the aluminum thin film deposited in the previous lab by photolithography and wet etching.

THEORY:

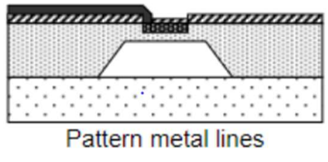
Aluminum, with a density of 2.7 g/cm3, is among light metals. Its crystal structure is cubic, specifically face-centered cubic. Its excellent electrical conductivity makes it a preferred choice for metallization in integrated circuits.

We have seen etching of Si and SiO_2 in previous labs, now we will see etching of metal. Aluminum is etched to pattern the wiring of the pressure sensor. One wet etchant of Aluminum is Phosphoric+ Acetic + Nitric acid, and photoresist can be used as a mask. The etchant used in this lab is Al etch 80-15-3-2, and its composition is Phosphoric acid (80%) + Water (15%) + Acetic acid (3%) + Nitric acid (2%), this etchant is also known as PAA etchant. The etching profile is more isotropic.

Here is a breakdown of the reactions involved:

Water and Aluminum reacts to form $Al(OH)_3$ and H_2 gas. The phosphoric, acetic, and nitric acid dissociate in water to form H^+ ions. The H^+ ions perform dissolution of Al ions. The H^+ ions react with Al surface forming $AlPO_4$ and $Al(CH_3CO_2)_3$ compound. These soluble compounds are dissolved and etches the Al. Nitric acid also acts as an oxidizing agent, oxidizing Al to form Al ions while it is reduced to NO_2 gas. The temperature at which the etching is carried also determines the etch rate.

The photolithography described in lab 2 is carried out the same way except for some changes. While Spin-Coating the Photoresist, HMDS is not used as an adhesion. The photoresist is directly deposited, and spin-coating is performed. The mask used for the photolithography is a clear field mask unlike those used in previous labs, which are all dark field masks.



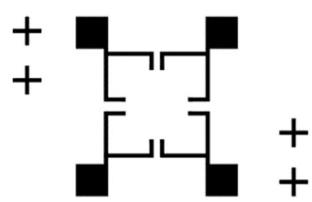


Figure: 37 & 38: Metal pattern masking

PROCEDURE:

-Proceed with the photolithography steps described in Lab 2. The only difference is HMDS is not used here.

-Alignment of mask and microscope first and then aligning wafer and the mask.

-Al etchant (Al etch 80-15-3-2) is heated up to 40°C in a glass beaker. The etch rate at this temperature is 600 nm/min with stirring and agitation. The rate increases by double with every 10°C increase in temperature

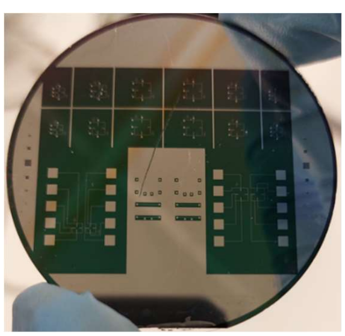
-The wafer is immersed in the etchant and agitated mildly, as the Al gets etched off it is observed that the wafer surface starts turning darker and pattern begins to form.

Figure 39 & 40: Aligning mask & wafer

Figure 41: Al etching

-Etching is carried out for 2 min and 50 s, wafer is rinsed and dried with N2 gun.

-The wafer is inspected under the optical microscope to make sure there is no unetched Al over the surface.



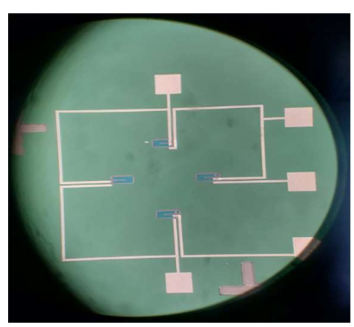


Figure 42: After etching & PR removal

Figure 43: Microscopic View

Theoretical etch rate @40°C = 600 nm/min

Al thin film thickness = 0.5 μm

Etch time @40°C = $\frac{0.5\mu m}{600\ nm}$ = $\frac{0.5\times10^{-6}}{600\times10^{-9}}$ = 0.833 min = 50 s

OBSERVATION:

There is a mismatch of 2 minutes between the etch time calculated theoretically and the etch time carried out during the lab. Although the etching was carried out under same conditions, the mismatch is due to some of these reasons:

-Initial thickness of the Al film more than 0.5 μm.

-The presence of thin layer of Al_2O_3 on the surface, needing it to be etched through before bulk Al is etched could also slow down.

-Variations in agitation intensity and minor variation in temperature.

-Impurities in etchant solution may cause local etch rate variation in the Al surface

During the mask wafer alignment, misalignment between the boron doped PZR and vias is observed. Misalignment observed between PZR, vias and metal line. Both resulted due to improper alignment during photolithography process.

Figure 44: Misalignment

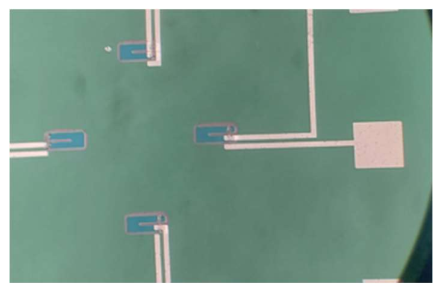


Figure 45: Misalignment after etching

LAB 12: SHEET RESISTANCE MEASUREMENT AND PACKAGING

AIM:

Measure the sheet resistance of Al sheet and PZR sheet by using 4-point probe and packaging and wire bonding of the sensor chip

THEORY:

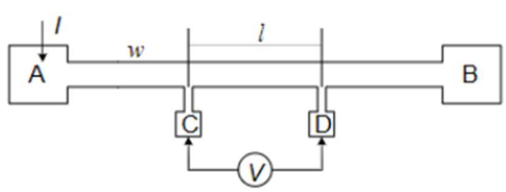
Sheet resistance can be defined as a measure of the resistance of a thin uniform film, per unit square. It's expressed in Ohms per square (Ω/□) It depends on the resistivity of the material and the thickness of the film. It characterizes how well a material conducts electricity across its surface and is often used in the semiconductor industry to describe resistance of conductive layers in integrated circuits or electronic devices. For a given diffusion profile, sheet resistance is uniquely related to the surface concentration (NS) of the diffused layer and the background substrate concentration of the wafer (NB); the latter affects the junction depth or t.

L/W = no. Of unit squares in the resistor

R= $\rho \frac{L}{Area} = \frac{\rho \times L}{w \times t} = \frac{\rho}{t} \times \frac{L}{w} = Rs\frac{L}{w}$, where Area = Width x thickness, $\frac{\rho}{t}$= Rs

$$Rs\frac{L}{w} = \frac{V}{I} \Rightarrow Rs = \left(\frac{V}{I}\right) / \left(\frac{L}{w}\right)$$

We used a 4-point probe to measure the sheet resistance of Al sheet and PZR sheet. The setup consists of 4 evenly spaced electrodes, in the form of probes, which are brought to contact the surface of material. A known constant current is applied through the outer two probes (A and B). Inner two probes (C and D) measure voltage across a small, localized area.



where I is the distance between two inner probes. The measurement is not affected by non-ohmic contact effects or other poor contact problems as there is no current flowing through the inner probes.

PROCEDURE:

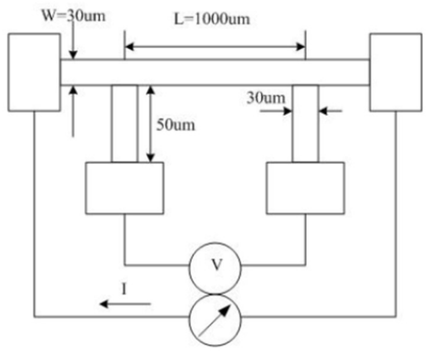


Figure 46: Setup is shown

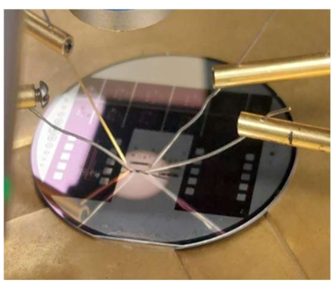
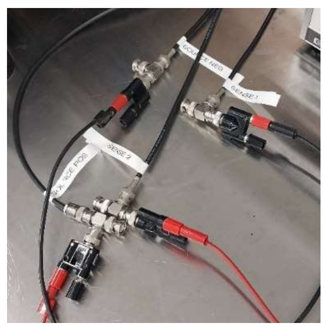
  

Figure 47, 48 & 49: Connections and from top- Multimeter, Ammeter, Current Source

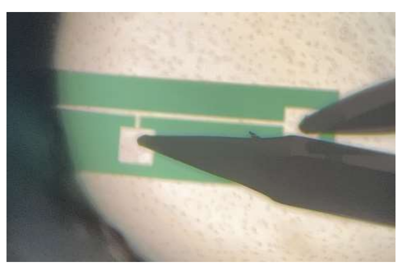


Figure 50: Probe in Contact

-The outer two probes are connected to the current source. The inner two probes are connected to the multimeter. The ammeter is connected to the current source.

-The probe is contacted with the Al line on wafer carefully to avoid scratching.

-5 readings taken, current increased by step of 1 mA from 1 mA till 5 mA, corresponding voltage measured.

-The probe is contacted with PZR line, and 5 readings taken, with current increased by step of 0.1 mA from 0.2 mA till 0.7 mA.

READINGS:

Current (mA)	Voltage (mV)	V/I	Avg V/I
1	4.22	4.22	4.2022
2	8.39	4.195	
3	12.56	4.186	
4	16.76	4.19	
5	21.1	4.22	

Table: Voltage Measurement of Al Sheet

Current (mA)	Voltage (V)	V/I	Avg V/I
0.2	1.43	7150	7164.9
0.3	2.15	7166.6	
0.4	2.88	7200	
0.5	3.59	7180	
0.6	4.29	7150	
0.7	5	7142.8	

Table: Voltage Measurement of PZR Sheet

CALCULATIONS:

For bulk Al, theoretical ρ = 2.82 × 10-8 Ω·m, t = 0.5 μm, Rs = ρ/t = 5.64×10-2 Ω/□ = 0.0564Ω/□

From readings, V/I = 4.2022 Ω and, l= 1000 μm and w= 30 μm.

$$Rs = \frac{V}{I} / \frac{L}{W} = 4.2022 \times \frac{30\times10^{-6}}{1000\times10^{-6}} = 0.126066\ \Omega/\square$$

For PZR, theoretical value,

PZR sheet resistance from reading, V/I =7164.9, l =1000 μm and w= 30μm

$$Rs = \frac{V}{I} / \frac{L}{w} = 7164.9 \times \frac{30\times10^{-6}}{1000\times10^{-6}} = 214.947\ \Omega/\square$$

OBSERVATION:

The Al sheet resistance calculated theoretically is different from the sheet resistance obtained from the readings. This is due to the presence of higher contact resistance between the probe and the material, which lead to higher sheet resistance and theoretical calculation of Al sheet resistance assumes perfectly homogeneous material with uniformity. In practical terms the material can have differences in doping concentration, crystalline structure, or other properties that can affect the electrical conductivity and result in deviations and also due to misalignment of probe spacing.

LAB 13: DEVICE TESTING

AIM:

Test the fabricated pressure sensor chip by applying pressure in a pressure chamber.

THEORY:

A piezoresistive pressure sensor consists of a thin diaphragm, and piezo resistors formed by doping of Si. The diaphragm is usually very thin and flexible to allow it to deform in response to applied pressure. When pressure is applied to the diaphragm of the sensor, it deforms, causing mechanical stress or strain in the piezoresistive material. This stress or strain leads to a change in the resistance of the piezoresistive material. The change in resistance of the piezoresistive material is measured using an electrical circuit connected to the sensor. This change in resistance is proportional to the applied pressure.

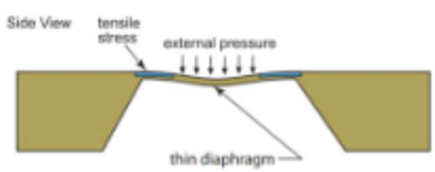


Figure 51: Diaphragm Under Pressure

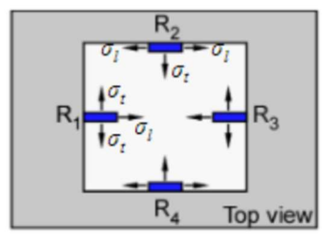


Figure 52: Piezo resistors on Diaphragm

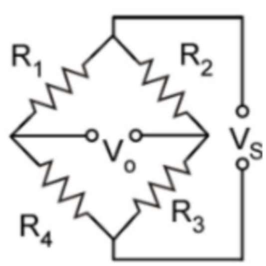


Figure 53: Wheatstone Bridge Config of PZR

The points of maximum stress are situated at the midpoint of each edge of the anchor adjacent to the diaphragm. To achieve the most notable change in resistance, it is advisable to position the piezo resistors in the midpoint of each edge of anchor.

In a diaphragm piezo resistors experience both longitudinal and transversal stress. The change in resistance is given by $\frac{\Delta R}{R} = \sigma l\, \pi l + \sigma t\, \pi t$, where $\pi t\, and\, \pi l$ are transversal and longitudinal piezo resistive coefficients.

The symmetric nature of the diaphragm ensures that similar stress occurs on both sides. This arrangement doubles the resistance change, thus enhancing sensitivity. For resistance readout, the piezo resistors are connected in the Wheatstone bridge configuration, where two resistors increase with increasing pressure while the other two resistors decrease. Longitudinal dominant R1 and R3 increase as pressure increases and transversal dominant R2 and R4 decreases.

The output voltage Vo is given by, $\frac{Vo}{Vs} = \frac{R1R3 - R2R4}{(R1+R4)(R2+R3)}$, where Vs is input voltage.

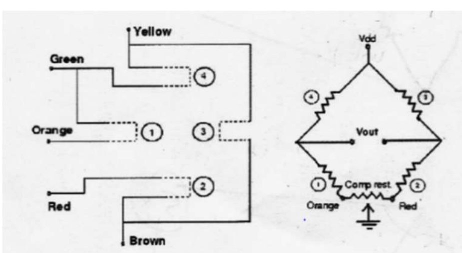
When resistance value changes, $\alpha 1 = \frac{\Delta R1}{Ro} = \frac{\Delta R3}{Ro} \quad \alpha 2 = \frac{\Delta R2}{Ro} = \frac{\Delta R4}{Ro}$,

$R1 = R3 = (1 + \alpha 1)Ro \quad R2 = R4 = (1 - \alpha 2)Ro$, rewriting $\frac{Vo}{Vs} = \frac{\alpha 1 + \alpha 2}{2 + \alpha 1 - \alpha 2}$.

Distinct advantages of using Wheatstone bridge configuration are: Converts the resistance change directly to a voltage signal for accurate readout. Low temperature influences. Ideally with balanced bridge, resistance changes equally due to temperature, so the output of bridge remains zero. $Vo = \Delta V = \frac{\Delta R}{R} Vs$, Rs = 0 stress resistance.

PROCEDURE:

First, place the pressure sensor in pressure chamber, and measure the resistance of individual piezo resistors by varying the pressure from 0 to 30 psi increasing by an interval of 5psi. After reaching 30 psi, decreases back to 0 by same interval.



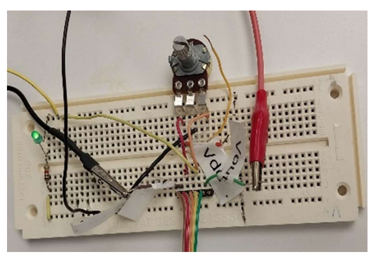


Figure 53 & 54: PZR Connections

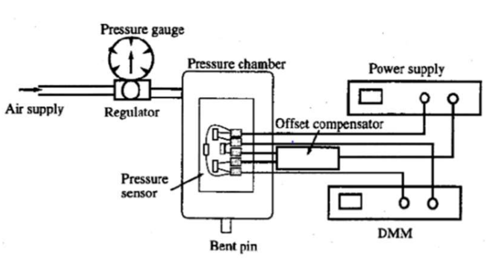


Figure 55 & 56: Pressure Chamber

-To balance the two sides, add compensation resistor in between orange and red.

-Tune compensator till offset voltage is 0V.

-Apply pressure from 0 to 30 psi and back from 30 psi to 0 psi in interval of 5 psi and measure the output voltage across green and brown at each pressure value.

READINGS:

Pressure (psi)	R1 (kΩ)	R2 (kΩ)	R3 (kΩ)	R4 (kΩ)	Voltage (mV)
0	1.53	1.5297	1.5387	1.5159	0
5	1.54	1.516	1.5454	1.5069	33.45
10	1.55	1.5108	1.5489	1.5010	55.75
15	1.5539	1.5052	1.5535	1.4946	75.91
20	1.5589	1.500	1.5579	1.4883	97.66
25	1.5636	1.4945	1.5624	1.4825	119.82
30	1.5691	1.4883	1.5677	1.4757	145.97
25	1.5631	1.4947	1.5621	1.4827	118.43
20	1.5579	1.5003	1.5571	1.4892	94.57
15	1.5534	1.5061	1.5525	1.4951	72.86
10	1.5487	1.5107	1.5485	1.5005	47.63
5	1.5451	1.5156	1.5449	1.5069	35.18
0	1.5382	1.5244	1.5388	1.5158	0

CALCULATIONS:

1. Pressure Range – 0, 5, 10, 15, 20, 25, 30 psi. (0-30psi)
2. Pressure Sensitivity - S = Normalized change in output signal per unit change in applied pressure

$$S = \frac{\Delta V}{Vs} \times \frac{1}{\Delta p} = \frac{145.97\times10^{-3}}{5\times30} = 0.9731 \text{ (mV/V/psi)}$$

$$\text{or } \frac{145.97\ mV}{30\ psi} = 4.86 \text{ mV/psi}$$

3. Offset:

   Initial offset voltage = 43.704 mV

   Compensated offset using compensation resistor = 0

4. Span (mV) = Device output signal over pressure range, i.e., Full scale output minus offset (mV)
   Output at 30 psi = 145.97 mV, output at 0 psi = 0 mV, (after compensation)

Span = 145.97 - 0 = 145.97 mV

5. Hysteresis

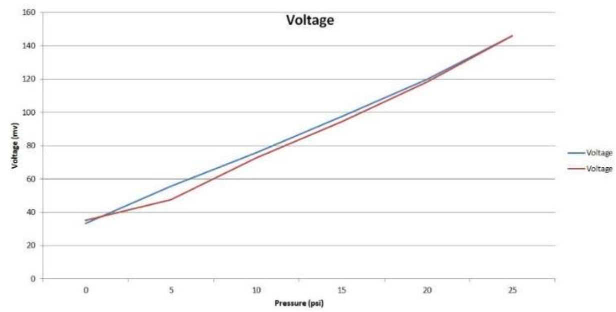


Figure 57: Voltage vs Pressure

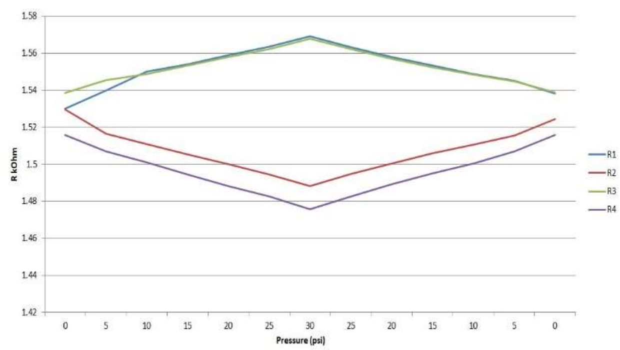


Figure 58: Resistance vs Pressure

6. Input Impedance:

   Measure input impedance = 2.0375 kΩ

7. Linearity:

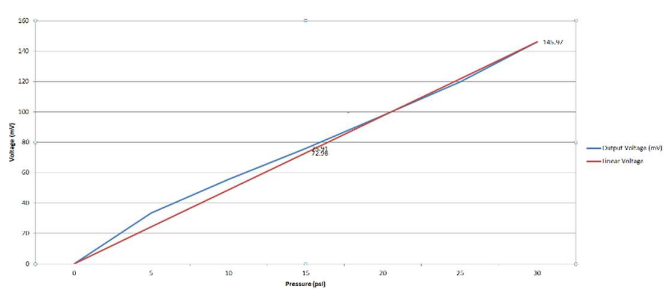


Figure 59: Terminal Based Linearity

V_1= 0 mV; V_2 = 75.91 mV; V_L= 75.91 - 72.98 (mV) = 2.93 mV; $V3$= 145.97 mV

Linearity % FS = (V_L/ (V_3 –V_1)) x 100 = $\frac{2.93}{(145.97-0)} \times 100 = 2.007$ %